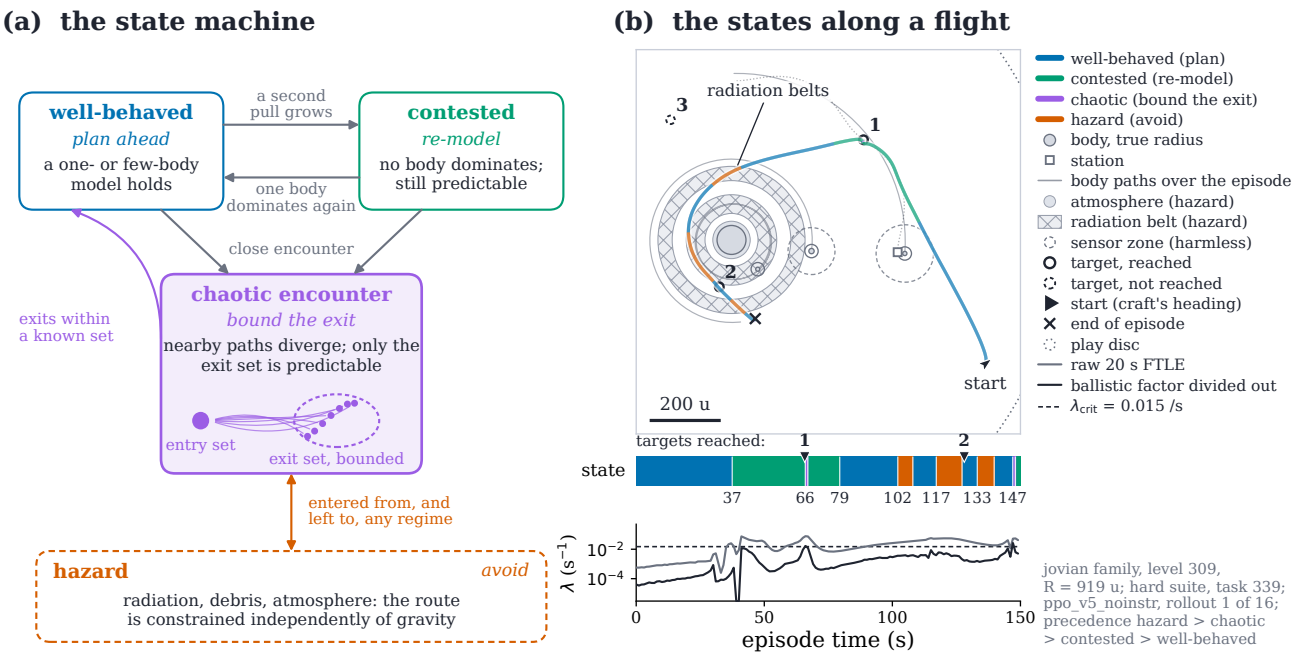

(a) the state machine
well-behaved
plan ahead
a one- or few-body model holds
a second pull grows
contested
re-model
no body dominates; still predictable
one body dominates again
close encounter
chaotic encounter
bound the exit
nearby paths diverge; only the exit set is predictable
entry set
exit set, bounded
exits within a known set
entered from, and left to, any regime
hazard
avoid
radiation, debris, atmosphere: the route is constrained independently of gravity
(b) the states along a flight
radiation belts
1
2
3
start
200 u
targets reached:
1
2
state
37
66
79
102
117
133
147
λ (s⁻¹)
10⁻²
10⁻⁴
0
50
100
150
episode time (s)
well-behaved (plan)
contested (re-model)
chaotic (bound the exit)
hazard (avoid)
body, true radius
station
body paths over the episode
atmosphere (hazard)
radiation belt (hazard)
sensor zone (harmless)
target, reached
target, not reached
start (craft's heading)
end of episode
play disc
raw 20 s FTLE
ballistic factor divided out
λcrit = 0.015 /s
jovian family, level 309,
R = 919 u; hard suite, task 339;
ppo_v5_noinstr, rollout 1 of 16;
precedence hazard > chaotic
> contested > well-behaved